# Practice workbook · Pigpen cipher

Using a Pigpen grid to turn letters into symbols or symbols into letters, classifying it as substitution, and judging whether a grid is a valid Pigpen key.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-21

# 1. Question 4 · part b · 3 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

**4** Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

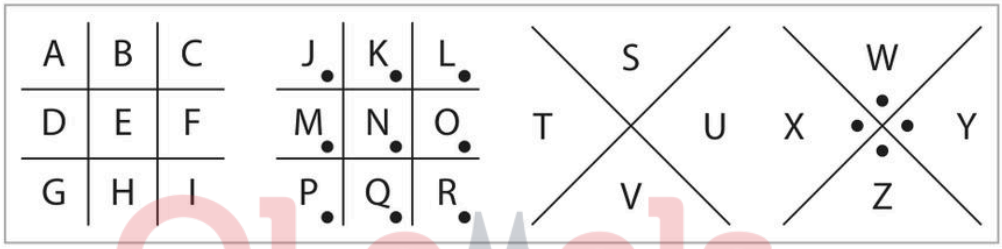



**Figure 1**

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

| **Letter** | M | A | Y |
|---|---|---|---|
| **Symbol** | | | |

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

**Figure 2** shows the first stage of the encryption process.

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| T | | | | | | | | I | | | | | |
| | H | | | | | | Y | | S | | | | |
| | | E | | | | M | | | | N | | | |
| | | | E | | E | | | | | | E | | R |
| | | | | N | | | | | | | | A | |

**Figure 2**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



P 7 2 4 0 6 A 0 8 1 6

## 1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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## 2. Question 4 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

**4** Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

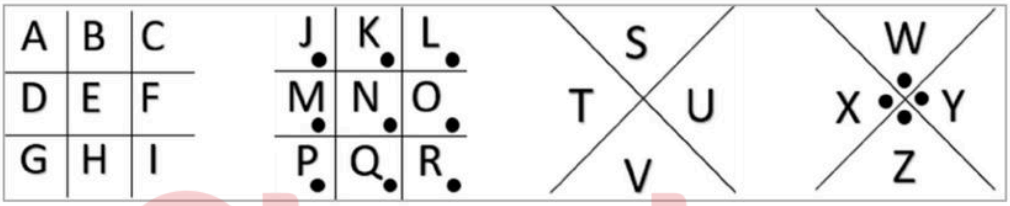



**Figure 1**

Complete the table to decode the symbols using the grid in **Figure 1**.

| Symbol | Letter |
|---|---|
|  | |
|  | |
|  | |
|  | |
|  | |
|  | |

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10

P 6 1 8 8 1 R A 0 1 0 1 6

## 2. Question 4 · part a · 1 mark · Working and answer

Working and answer

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## 3. Question 4 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

**4** Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

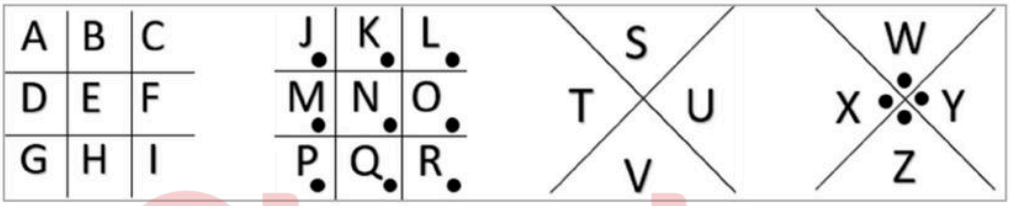



**Figure 1**

Complete the table to decode the symbols using the grid in **Figure 1**.

| Symbol | Letter |
|---|---|
|  | |
|  | |
|  | |
|  | |
|  | |
|  | |

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10

P 6 1 8 8 1 R A 0 1 0 1 6

## 3. Question 4 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) **Figure 2** shows another grid.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| C | O | M | A | B | D | L | N | Q |
| P | U | T | F | G | H | V | W | X |
| E | R | S | I | J | K | Y | Z | |

**Figure 2**

State whether this set of symbols meets the requirements of a Pigpen cipher.

Justify your answer.

(3)

State

Justification

**(Total for Question 4 = 7 marks)**

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## 3. Question 4 · part b · 3 marks · Working and answer

Working and answer

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# 4. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

**4** Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

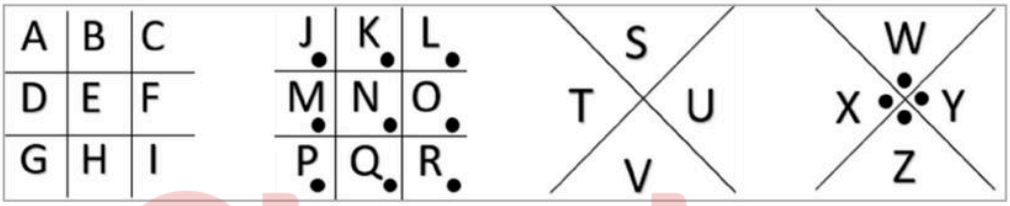



**Figure 1**

Complete the table to decode the symbols using the grid in **Figure 1**.

| Symbol | Letter |
|---|---|
|  | |
|  | |
|  | |
|  | |
|  | |
|  | |

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

# 4. Question 4 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) **Figure 2** shows another grid.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| C | O | M | A | B | D | L | N | Q |
| P | U | T | F | G | H | V | W | X |
| E | R | S | I | J | K | Y | Z | |

**Figure 2**

State whether this set of symbols meets the requirements of a Pigpen cipher.

Justify your answer.

**(3)**

State

.....

Justification

.....

.....

.....

.....

**(Total for Question 4 = 7 marks)**

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## 4. Question 4 · part c · 3 marks · Working and answer

Working and answer

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## 5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

## 5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

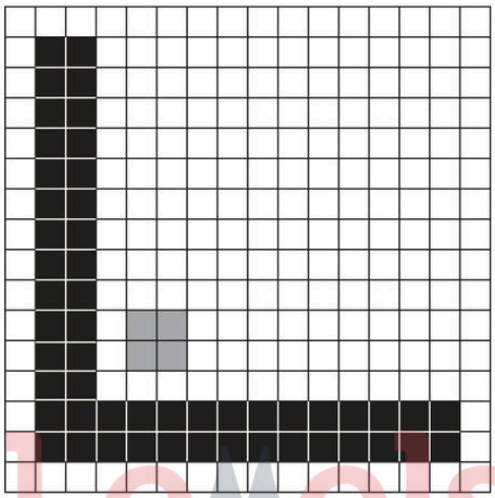


**Figure 1**

(i) State the smallest component of a bitmap image.

**(1)**

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

**(4)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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## 5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

**(2)**

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

THIS AREA

(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

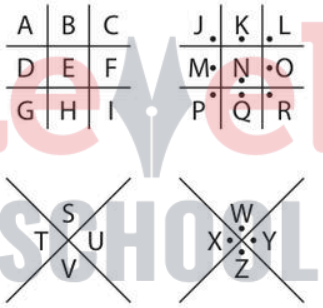
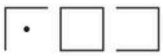



**Figure 2**

Complete the table to give the decrypted plain text.

**(2)**

| Encrypted text | Plain text |
|---|---|
|  | |
|  | |

**(Total for Question 3 = 14 marks)**

## 5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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